

Precision and Recall

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Calculate both ratios for Precision recall and denominators. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Precision recall and denominators. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Use $[[8, 2], [3, 6]]$. Give TP, FP, FN, precision, recall and accuracy.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

TP=9, FP=3, FN=6, TN=12. Compute precision and recall. What happens to precision when a system predicts no positives?

6. Solve the given case

Positive is spotted: TP=8, FP=2, FN=4.

Compute precision and recall.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

With disease positive, TP=6, FP=2 and FN=4. Calculate precision and recall, showing both denominators.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 17 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

TP=9,FP=3,FN=6,TN=12. Compute precision and recall. What happens to precision when a system predicts no positives?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Identify the groups

For the worked matrix, give predicted-positive and reference-positive totals.

My reasoning and evidence

13. W2 Calculate both ratios

Calculate precision, recall and accuracy for the worked matrix.

My reasoning and evidence

14. W3 Handle no flags

Explain precision and recall for $[[16, 0], [4, 0]]$.

My reasoning and evidence

15. W4 Apply independently

Use $[[8, 2], [3, 6]]$. Give TP, FP, FN, precision, recall and accuracy.

My reasoning and evidence

16. W5 Limit a claim

A team reports recall 100 percent but hides precision and counts. What is missing?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Precision and Recall

Put precision and recall denominators on the picture instead of memorising names.

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep precision and recall denominators distinct

	predicted spotted	predicted healthy
actual spotted	6	4
actual healthy	2	8

Read a cell using both its row and column labels.

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predicted positives

Eight leaves were flagged: $\text{precision} = 6/(6+2) = 75\%$.



2. Actual positives

Ten leaves were spotted: $\text{recall} = 6/(6+4) = 60\%$.



3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Put precision and recall denominators on the picture instead of memorising names.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

1. Predicted positives
2. Actual positives
3. Keep goals separate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Keep goals separate Precision asks how reliable flags are; recall asks how many spotted leaves were caught. Read the picture in order. Precision asks how reliable flags are; recall asks how many spotted leaves were caught. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Correct two false negatives into true positives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$. TP becomes 8 and FN 2; FP remains 2, so the two denominators happen to equal 10.

Repair a missing building block

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf predictions from training examples. Short practice: Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
